

GreenSight

Monitoring and Assessment of Vegetation Profile with AI, LiDAR Technology and Hyperspectral Satellite Imagery

Team Members	Faculty Guide	Exhibit Type
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Focus Areas: GeoAI & Machine Learning for Earth Analytics Drone/UAV Mapping & LiDAR Applications Environmental Monitoring.		

PROBLEM STATEMENT

India loses vast forest and agricultural land annually due to undetected vegetation stress, deforestation, and land degradation. Traditional manual field surveys are labour-intensive, costly, and impossible to scale across Karnataka's diverse ecosystems — from the Western Ghats to the Deccan Plateau. Current remote sensing tools analyse either LiDAR structural data or spectral satellite imagery independently, leaving critical gaps in comprehensive vegetation intelligence. This disconnected approach delays timely government intervention, makes crop advisory inaccurate, and hinders the State Forest Department's ability to assess biomass and ecological risk at scale.

PROPOSED SOLUTION & INNOVATION

Green Sight is a unified AI-driven vegetation monitoring framework that fuses LiDAR point cloud processing from (OpenTopography), hyperspectral satellite imagery (HySIS via ISRO's Bhoonidhi Portal), and deep learning into one automated pipeline — directly aligned with KRSRAC's geospatial mission.

Key innovations:

- LiDAR Pipeline: Noise removal → ground filtering → 3D canopy height model → tree count (39,000+ trees detected) and vegetation risk heat maps flagging trees >15 m**
- GeoAI Models: CNN, CHM and Random Forest for forest-type classification & Density. (Tropical forest 88.35%, Dry forest 25.05%, Temperate forest 14.21%) biomass estimation and more.
- Utilizes HySIS spectral soil classification and LiDAR data to analyze soil and terrain, recommending optimal Kharif/Rabi and other crops with growth timelines for accurate, location-specific agricultural decision-making for Karnataka/Indian farmers.
- Terrain Slope Analysis: Classifies flat vs. steep terrain to guide agricultural suitability and agroforestry planning
- Live Web Dashboard: All outputs — 3D models, heat maps, crop advisory — accessible at greensight-lidar.base44.app (worked demo available)

Scalability & Impact:

The framework is dataset-agnostic and scalable to any Indian region using LiDAR datasets and satellite data — no new hardware required. It can be directly integrated into KRSRAC's existing geospatial workflows to support Karnataka's State Forest Department, agricultural planning bodies, and disaster preparedness units, reducing survey costs by over 70% compared to traditional methods.